Applications of Virtual Reality in Computer Sciences Education: A Systematic Literature Review

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***Abstract*— Virtual Reality (VR) has become increasingly popular recently, presenting an exciting opportunity for its integration into Computer Science. VR offers an immersive and interactive learning environment specifically tailored for Computer Science education. To explore this potential, we conducted research using the Systematic Literature Review (SLR) method. Our Study focused on three key aspects: the devices utilized in VR, the different contexts in which VR is applied in Computer Science, and the evaluation methods used to assess the effectiveness of VR in this field.**

Keywords*— *Virtual Reality, Computer Science Education, Systematic Literature Review, VR

I. Introduction

Virtual Reality (VR) is a technology that originated in the 1970s and has gained widespread popularity in recent years. The term "Virtual Reality" consists of two components:" Virtual" and "Reality". "Virtual" refers to a state or condition that exists on the internet or within the realm of technology, while "Reality" denotes the state of an actual event that is real. Thus, everything that takes place within Virtual Reality is a genuine experience but happened virtually or within the digital world[1][2].

Virtual Reality (VR) has become increasingly popular among teenagers to engage their sociality. These kinds of immersive spaces have captured the attention of teenagers through various means, such as immersive gaming, virtual intimacy, and emotional connections [3]. While VR has the potential for rich social engagement, it also poses a potential risk for teenagers. Therefore, it is crucial that through empirical research to understand how teenagers perceive and engage with VR, this will provide a valuable insight into their unique interaction with these new online social spaces.

To access Virtual Reality, it is necessary to have the appropriate equipment and software. This may include a VR headset or goggles, and software that enables interaction with virtual environments. Once the required equipment and software have been obtained and installed, the user can put on the headset or device to begin interacting with the virtual environment. This could involve engaging in gaming activities, exploring virtual spaces, or participating in a range of other immersive VR experiences.[4]

Due to the COVID-19 outbreaks in 2020, students were required to attend remote learning through online platforms. Many schools and institutions used various conference tools such as Zoom, Google Meet, and Microsoft Teams. However, not all students were able to remain attentive during these online learning sessions, and most interactions were one-way conversations. Despite many students joining video conferences, some of them did not pay attention to the teacher and instead doing other activities.[5]

In recent years, Virtual Reality has been used for entertainment or games. Since Covid-19 comes, Virtual Reality is also used for online learning which is examined in one study by Cecep Kustandi (2019). This study discovered that virtual reality brings many benefits, improving retention, and experimental learning. Another study by Enes Ygitbas (2022), Virtual Reality can be effective as an alternative for teaching and group work, as it enhances student engagement and motivation throughout the learning process. In the third study by Stelian Nicola, Lacramioara Stoicu-Tivadar, And Alexandru Patrascoiu (2018), The study discovered that Using VR applications among students has led to a more effective learning of Bubble Sort Algorithms.

This research aimed to investigate the most used devices and devices type for virtual reality (VR) technology in the computer science field. Additionally, we sought to gain insight into the purposes of students utilizing VR technology in Computer Science, and to find how to evaluate VR applications withing this field.

To acquire a deeper grasp of virtual reality for learning, a literature review is required. SLRs are a standard strategy for finding answers to research questions by performing a literature review of past relevant studies. To reach a thorough outcome, our study used multiple popular database journals.

In this paper, we also explore Systematic Literature Review (SLR) through the implementation of virtual reality (VR) in computer science. This is because VR implementation has been trending, as evidenced by its increasing quantity[12,20]. Furthermore, in the context of implementing VR in the field of computer science, it assists researchers or VR developers in understanding which devices are most commonly used, how VR applications take shape in computer science, and how application evaluations are conducted.

The paper will be structured into three distinct sections. Section 2 will be the outline of the research methodology employed, specifically focusing on the systematic literature review utilized in this study. Section 3 will present the study's findings, and section 4 will be the conclusion of this paper.

II. Literature Review

A. Systematic Literature Review

The research method used in this study is a systematic literature review (SLR). SLR is a comprehensive, transparent, and reproducible way of synthesizing scientific evidence. It seeks to include all published evidence on a topic and appraise the quality of this evidence. SLRs identify, select, and critically appraise research to answer a clearly formulated question.

This research leverages a variety of well-established sources such as IEEE Xplore, Science Direct, ACM Digital Library, ResearchGate and Mendeley to provide comprehensive results. The SLR process includes defining the research scope and research questions, collecting research data, and analyzing and summarizing the research [6].

B. Related Works

A few SLR studies have been carried out to investigate topics associated with VR and offer suggestions for its creation and application. Virtual Reality in Computer Science Education: A Systematic Review was one of the SLR researches carried out by Johanna Pirker and colleagues. In this publication, the potential, and applications of virtual reality (VR) for computer science education. To integrate virtual reality (VR) technology into computer science education, the essay conducts a thorough examination of the literature in the field. The main emphasis is on identifying key elements including learning objectives, employed technologies, interaction characteristics, as well as the benefits and challenges associated with the use of VR in computer science education.[7] Another SLR study conducted by Yummeng Han was Literature Review: Virtual Reality in Engineering Education. This study uses a literature review to analyze the concept, characteristics, and research on the teaching of virtual reality technology. Virtual reality technology is developing rapidly, and it is changing the way people think, remember, and educate their children. [8] On the other hand, the study conducted by Cecep Kustandi was Virtual Reality Use in Online Learning. This study developed the learning of virtual reality objects used in online learning. The suitability of the utilization of pictures, materials, videos, and supporters within the learning model has a positive impact on students. Both quantitative and qualitative studies have revealed that this learning model has the ability to enhance students' motivation to learn and improve their learning skills [9]. Additionally, a research paper titled "Lord of Secure: The Virtual Reality Game for Educating Network Security" introduces a VR game that can be used to teach network security concepts effectively. The game, called "Lord of Secure", is available on Android and offers an alternative learning experience for students. The game aims to foster students' understanding of network security concepts beyond what traditional classroom settings can provide [10].

III. Research METHODOLOGY

In this paper, we implemented data presentation similar to a previous study on SLR in Virtual Reality [59], addressing comprehensive presentation using tables.

A. Research Question

The research question guides the literature review by helping identify relevant data. Table 1 shows the research questions of this study.

TABLE I. Research Question

ID	Research Question	Motivation
RQ1	What devices and types are used for experiencing Virtual Reality?	To determine devices used when experiencing virtual reality
RQ2	What are the contexts of using virtual reality in computer science?	To determine the context of using virtual reality in computer science
RQ3	How to evaluate computer science related virtual reality apps?	To determine what evaluation used in computer science related virtual reality app

B. Research Finding

To answer the stated Research Questions, specific terms looked through research papers published in various well-known sources. The search term utilized was "Virtual Reality" AND "Computer Science", another search term utilized was "Virtual Reality Device Used," and the last search term was "Virtual Reality User Experience Evaluation".

TABLE II. Search Result of Each Database Journal

No	Database Journal	Number Of Article
1	ResearchGate	1000
2	Mendeley	900
3	Science Direct	7466
4	IEEE Xplore	6223
5	ACM	6598
	Total	22187

TABLE III. Inclusion and Exclusion Criteria

Criteria		
Inclusion	***I1***	Articles that discuss the purposes of using virtual reality (VR) in computer science, the devices used, and evaluation
	I2	Articles published after 2018.
	I3	Articles in the English language
Exclusion	***E1***	Articles that are equivalent in content, but published in different journals

Criteria		
	E2	Articles that are not related to the use of VR in computer science and evaluation.
	E3	Articles that do not discuss the devices used.

IV. Result and Discussion

A. Device and Type Used

Virtual reality is frequently used to educate and learn engineering, particularly in computer science education [11]. Many publications have been analyzed, and it has been discovered that virtual reality is usually classified as either completely fully immersive or non-immersive [12, 13].

TABLE IV. Virtual reality immersive types

Immersive Type	Ref
Fully Immersive	[12], [27],[28]
Non-Immersive	[13], [29]

To fully immerse ourselves in virtual reality, we need goggles to make us feel more into the virtual environment and separate us from real life surroundings [14]. These goggles are called "Head Mounted Display" (HMD). It is shown that HMD is used to experience virtual reality in many papers [15, 16, 17, 18]. HMD can be used for smartphone, for example HMD that is used for smartphone are Samsung Gear VR, Google Daydream View, and Google Cardboard [19, 20, 21, 22], and then there is HMD that are completely not needing any cables or external device, which is standalone HMD, Oculus Go, HTC Vive, and Oculus Quest are the examples of standalone HMD [23], [24], [25], [26], lastly HMD also can be used for PC/Desktop, example of HMD for PC is Oculus Rift[52].

TABLE V. Device types and used

Device Types	Device Used	Ref
Smartphone Based VR	Samsung Gear VR, Google Daydream View, Google Cardboard	[31], [32]. [33], [34]
Standalone VR	Oculus Go, HTC Vive, Oculus Quest	[35], [36], [37], [38]
PC Based VR	Oculus Rift S	[52]

B. Context of using Virtual Reality in Computer Science Education

Virtual Reality (VR) offers many advantages in Computer Science Education (CSE). One way VR is used in CSE is for studying programming. In a reviewed paper, researchers used VR to teach the concepts of data structures like stack and queue [39]. Another example is a cloud-based VR platform called cr3d.io, which aims to make coding classes more fun and engaging compared to traditional methods. Using visual coding blocks, c3d.io allows students to learn coding by creating animations or games [40]. Another VR platform, "My Reality," helps students learn programming concepts in their own environment [42]. Additionally, there is a VR game designed to make programming concepts easier to understand. This game teaches variables, lists, and arrays in a more accessible and enjoyable way for learners [41].

Virtual Reality is also used for Computational Thinking, and an excellent context is a VR game called "IThinkSmart." This game incorporates three virtual mini games: River Crossing, Tower of Hanoi, and Mount Patti treasure hunt. By combining these games, IThinkSmart creates an immersive, interactive, engaging, and personalized learning experience, making the learning process more enjoyable and effective [43].

Virtual reality is also being utilized in the realm of security research. One notable example is a game called "VRFIWALL," which aims to teach computer science students about various aspects of firewalls, including their definitions, types, mechanisms, capabilities, and limitations [44]. Another game called "Lord of Secure" takes players into a virtual world where they can explore the intricacies of network security. To reinforce learning, this game also includes quizzes to test users' after learning knowledge [45].

Cloud computing has revolutionized the field of computer science, and recent studies have explored its potential in various areas. One such study focuses on Mobile Edge Cloud, an advanced version of ACM (Adaptive Cloud Migration), which leverages a virtual reality gaming application to facilitate live migration of cloud services across multiple hosts worldwide [46]. This groundbreaking approach allows for seamless service transfer and enhances the overall gaming experience. In addition, the modern educational landscape is experiencing rapid transformations driven by technology. Artificial Intelligence (AI), machine learning, cloud computing, and virtual reality have emerged as key drivers, offering novel avenues for digitizing and modernizing education to cater to the needs of the current generation of learners [47].

In various areas of Computer Science research on IoT, Students can actively take advantage of the learnIoTVR learning environment, an end-to-end Virtual Reality learning environment that can help students to acquire IoT knowledge through immersive design, programming, and exploration of real-world environments empowered by IoT [48]. Additionally, VR has proven valuable in providing therapy sessions for individuals with phobias, enabling them to receive expert guidance within a shared virtual reality environment [49]. Furthermore, the combination of Virtual Reality, Augmented Reality, and IoT has paved the way for innovative solutions in physical rehabilitation. Notably, the development of serious games using Unity and technologies such as Meta Quest2 as an MX Interface has shown promising results [50].

Moreover, Virtual Reality-based Learning Environments (VRLEs) have gained significant popularity, thanks to the widespread availability of cloud infrastructure, edge technologies, and high-speed networks. This popularity has led to a growing need for incorporating IoT-based application design concepts within social VRLEs. By doing so, these environments can provide more scalable and efficient services that effectively adapt to the dynamic conditions of cloud/fog systems [51].

TABLE VI. VIRTUAL REALITY CONTEXT OF COMPUTER SCIENCE STUDY

Computer Science Concept	Context	Ref
Programming	Student can learn stack and queue; student can learn coding by making games using visual block code; student can learn variables, list, and arrays	[39], [40], [41], [42]
Security	Student can learn firewall concept; student can learn network security such as Firewall, IDS, IPS and Honey Pot	[44], [45]
Cloud Computing	Student can learn mobile edge cloud, Student can use AI, machine learning, cloud computing, and Virtual Reality,	[46],[47]
IOT	Student can LearnIoTVR, Tactile Internet Application, Rehabilitation using IOT, Using IoT in VRLE Social	[48], [49], [50], [51]

C. How to Evaluate Computer Science's related Virtual Reality Applications

a) User Interface/User Experience Questionnaire

There is a paper that assesses the virtual reality (VR) user interface (UI) and user experience (UX). The article specifically evaluates the UI/UX of a VR game built for network security education. The evaluation methodology included 28 students who were asked about their experience with the game's teaching technique and whether they felt they learned the information better than they would in a regular classroom setting. The findings revealed that over 90% of the students responded positively to the teaching style, showing a high level of comprehension. Furthermore, 82% of students believed they understood the topic better in the VR game than in the classroom [10]. Not only that, but there are also more papers using UIQ/UEQ [53, 58].

b) Usability Questionnaire

Another Paper "Usability and Learning Effectiveness of Game-Themed Instructional (GTI) Module for Teaching Stacks and Queues" This paper aimed to collect the students' demographic information and asked the students how they felt about the module's usability, immersion, and learning effectiveness[54].

c) Theory of Reasoned Action Survey (TRA)

TRA is used to evaluate a paper named "Educational Game-Theme Based Instructional Module for Teaching Introductory Programming" to examine the likability of GTI for learning linked list and binary tree. According to this paper's survey results, 22.64% of students say the GTI module is extremely interesting, while 50.94% think it is interesting[56].

TABLE VII. . EVALUATION METHODS

Evaluation Method	Ref
User Experience Questionnaire (UEQ)	[10], [53], [58]
Usability Questionnaire	[54], [55], [56], [57]
Theory of Reasoned Action Survey	[56]

V. CONCLUSION AND FUTURE WORKS

Virtual reality has become a valuable asset in the area of computer science education, providing students with captivating and immersive opportunities where the applications span across multiple areas of computer science education, such as Programming, Security, Cloud Computing, IoT, and Computational Thinking. The primary tool employed in virtual reality is Head Mounted Display, together with Oculus can bring the most used device for educational purposes.

For further enhance the user experience and deepen the level of immersion within virtual reality environments, future research endeavors may concentrate on refining interaction techniques, implementing haptic feedback systems, or advancing rendering technologies.

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